Screenshots

1 A

```
seed@VM: ~/.../PES1UG24CS269-pes-vcs
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed02
2 Wireshark b5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea
09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1 B

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ find .pes/objects -
type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5
e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1a
f8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650e
da471e566fcf
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
```

2 A

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2 B

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629
b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ xxd .pes/objects/25/ef1fa07ea68a5
2f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ xxd .pes/objects/2a/594d39232787f
ba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf | head -20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
```

3 A

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ make pes
make: 'pes' is up to date.
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ echo "hello" > file1.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ echo "world" > file2.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes add file1.txt file2.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.h
  untracked:  Makefile
  untracked:  index.h
  untracked:  tree.c
  untracked:  test_tree
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  commit.c
  untracked:  index.c
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  .gitignore
  untracked:  pes.c
  untracked:  test_tree.c
  untracked:  test_objects
  untracked:  commit.h
  untracked:  pes.h
```

3 B

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776779802 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776779804 6 file2.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
```

4 A

```
Committed: 8091692a400f... Initial commit
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ echo "World" >> hello.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes add hello.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes commit -m "Add world"
Committed: ef9380b0d12c... Add world
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ echo "Goodbye" > bye.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes add bye.txt
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes commit -m "Add farewell"
Committed: 104afbe1779B... Add farewell
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ ./pes log
commit 104afbe177988dfd685748b38c7cb6b41d3da7b957c0568c58f23cb5354c23f3
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776785089

    Add farewell

commit ef9380b0d12c23ccbaacd84a58fa7aa07191e06bbe054e19339136cfaa428c43
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776785089

    Add world

commit 8091692a400fe4adfe598e736fcb973c285dc0397bc6c6b35ffc98faf07f8fe2
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776785089

    Initial commit

[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
```

4 B

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/85/c8d777d0be85748b6788d1b5caf7d69754e6e68068c265ca21cbd6c85e5f43
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/26/24ef174525a6423abb03d4a38952e6c568db28abff2ec7a6dd5e933ef9b63a
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/46/5ee07f89e1236d53fd2c3c8c927b313d66ab5523b5888cee6988c7b2795316
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8807b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719686c36723cd16aefbc73b3dc8b74de61870183357d6968b9
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea89227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708258a89f315d4d8bb92703343c84df834438888e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/objects/fa/ee6b13211ad233380a92ec34fb636deb4ef88bd553a55b672683db740d6ebe
.pes/refs/heads/main
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
```

4 C

```
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ cat .pes/refs/heads/main
2624ef174525a6423abb03d4a38952e6c568db28abff2ec7a6dd5e933ef9b63a
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
```

Final

```
Untracked files:
  (nothing to show)


--- First Commit ---
Committed: ff222bef4e4f... Initial commit

Log after first commit:
commit ff222bef4e4f595c88e1311409e7208fb3a44eb8c3d45d036f91834cda598938
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776784582

    Initial commit


--- Second Commit ---
Committed: 63155d42001d... Update file.txt

--- Third Commit ---
Committed: c4933e530b9d... Add farewell

--- Full History ---
commit c4933e530b9dec1b925b547047e61710efc5c5124204b506430d3cb4b6e1ce32
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776784583

    Add farewell

commit 63155d42001ddde54b36124b77f8f3d3ab11070383249759e6e7e62282193117
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776784582
```

```
    Update file.txt

commit ff222bef4e4f595c88e1311409e7208fb3a44eb8c3d45d036f91834cda598938
Author: Meghna Sanjeev PES1UG24CS269
Date:   1776784582

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
c4933e530b9dec1b925b547047e61710efc5c5124204b506430d3cb4b6e1ce32

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5685f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e8d669579753 3f50d
.pes/objects/63/155d42001dddc54b36124b77f8f3d3ab11070383249759e6e7e62282193117
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c89635c5895ef7e68f7fa1974d981ce282f30e08de0ccd92
.pes/objects/c4/933e530b9dec1b925b547047e61710efc5c5124204b506430d3cb4b6e1ce32
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5358f9
.pes/objects/db/07a1451ca9544dbf66d769b585377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/ff/222bef4e4f595c88e1311409e7208fb3a44eb8c3d45d036f91834cda598938

=== All integration tests completed ===
[04/21/26]seed@VM:~/.../PES1UG24CS269-pes-vcs$
```

Branching and Checkout

Q5.1

To implement pes checkout <branch>, the system must first update the HEAD file so that it points to the selected branch. It then reads the commit hash stored in that branch and loads the corresponding commit object. From this commit, the root tree is obtained, which represents the complete snapshot of the project.

The working directory must then be updated to match this snapshot. This involves recreating all files and directories from the tree and removing any files that are not part of the target branch. Finally, the index must also be updated so that it reflects the state of the checked-out commit.

This operation is complex because it requires careful coordination between the working directory, index, and repository. It must also ensure that existing user changes are not accidentally overwritten.

Q5.2

A dirty working directory can be detected by comparing the current state of files with the state recorded in the index. For each tracked file, the system computes its current hash and compares it with the stored hash in the index. If they differ, the file has been modified.

To detect conflicts during checkout, the system also compares the index with the target branch's tree. If a file has been modified in the working directory and also differs between the current branch and the target branch, a conflict is detected. In such cases, checkout must be refused to prevent data loss.

Q5.3

In a detached HEAD state, the HEAD points directly to a commit instead of a branch. If new commits are made in this state, they are not associated with any branch and can become unreachable once the user switches to another branch.

To recover these commits, the user can create a new branch that points to the current commit. This ensures that the commits remain part of the repository history and are not lost.

Garbage Collection and Space Reclamation

Q6.1

To remove unreachable objects, the system must first identify all objects that are still reachable from existing branch references. This can be done by starting from each branch head and traversing the commit history, visiting all associated tree and blob objects.

All visited objects are marked as reachable. Any object in the object store that is not marked is considered unreachable and can be safely deleted. A set or hash-based structure can be used to efficiently track visited hashes.

For a repository with 100,000 commits and multiple branches, the traversal would visit all commits and their associated trees and blobs, resulting in a large but manageable number of objects proportional to the project size.

Q6.2

Running garbage collection concurrently with a commit operation is dangerous because of potential race conditions. For example, a commit may create new objects but has not yet updated the branch reference. At the same time, garbage collection may scan the repository and fail to see these new objects as reachable, leading to their deletion.

This can result in corrupted commits or loss of data. To prevent this, real systems like Git ensure that garbage collection does not run simultaneously with active operations. They use locking mechanisms and carefully designed procedures to ensure that newly created objects are not prematurely deleted.